

The whole thing in three moves

Every $\psi(t)$ question is this machine. Nothing else.

- Diagonalize H .** Get eigenpairs $(E_k, |E_k\rangle)$. The eigenvectors of H are the *energy eigenstates*; the eigenvalues are the energies.
- Expand the start state** in that basis: $|\psi(0)\rangle = \sum_k c_k |E_k\rangle$, $c_k = \langle E_k | \psi(0) \rangle$.
- Stamp a phase on each piece:**

$$|\psi(t)\rangle = \sum_k c_k e^{-iE_k t/\hbar} |E_k\rangle$$

Why it works: an energy eigenstate alone obeys $|\psi(t)\rangle = e^{-iEt/\hbar} |\psi(0)\rangle$ (plug into $\frac{d\psi}{dt} = -\frac{i}{\hbar} H\psi$ — lecture sign convention writes $\frac{i}{\hbar} H\psi$; the *solution* carries $e^{-iEt/\hbar}$, use that). Schrödinger is linear, so a superposition evolves piecewise.

Equivalent operator form

$$U_t = e^{-iHt/\hbar} = P e^{-iDt} P^\dagger, \quad H = P D P^\dagger,$$

$D = \text{diag}(E_k)$, P 's *columns* are the normalized eigenvectors, $e^{-iDt} = \text{diag}(e^{-iE_k t})$. Diagonal $H \Rightarrow$ just exponentiate the diagonal, no work.

Step 1 — eigen-decomposition of a 2×2

Eigenvalues

$$\det(H - \lambda I) = 0 \Rightarrow \lambda^2 - \text{tr}(H)\lambda + \det(H) = 0.$$

- A 2×2 has **two** eigenvalues (possibly equal). Never stop at one.
- Check: $\lambda_1 + \lambda_2 = \text{tr } H$, $\lambda_1 \lambda_2 = \det H$.
- $H - \lambda I$ subtracts λ from the *diagonal only*. For $\lambda = -1$ the diagonal entries **grow**: $a - (-1) = a + 1$.
- H Hermitian $\Rightarrow$ all E_k real, eigenvectors of distinct E_k orthogonal.

Eigenvectors — no row-reduction needed

For $H = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ and eigenvalue λ :

$$v = \begin{pmatrix} q \\ \lambda - p \end{pmatrix} \quad \text{or} \quad v = \begin{pmatrix} \lambda - s \\ r \end{pmatrix}$$

(use whichever is not the zero vector), then **normalize**.

- $(H - \lambda I)v = 0$ always has $v = 0$ as a solution. That is *never* the eigenvector — you need a nonzero one.
- The two rows of $H - \lambda I$ are proportional, so one row is enough. If the second row contradicts the first, λ is wrong.
- Verify in one line: $Hv \stackrel{?}{=} \lambda v$.

The shortcut this exam keeps rewarding

Almost every H on the practice tests is

$$H = \begin{pmatrix} a & b \\ b & a \end{pmatrix} = aI + bX.$$

Its eigenvectors are **always** $|+\rangle, |-\rangle$ — no determinant, no linear system:

$$H|+\rangle = (a+b)|+\rangle, \quad H|-\rangle = (a-b)|-\rangle.$$

H E_+ (on $|+\rangle$) E_- (on $|-\rangle$)

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	3/2	1/2
$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	3	1
$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$	2	0
$\begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}$	1	2

Note the last row: $b < 0$ flips which one is larger. Read b 's sign, don't assume $|+\rangle$ is the ground state.

Complex off-diagonal (same idea)

$\begin{pmatrix} a & be^{-i\varphi} \\ be^{i\varphi} & a \end{pmatrix}$ has $E = a \pm b$ with eigenvectors $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm e^{i\varphi} \end{pmatrix}$.

Step 2 — expanding $\psi(0)$ in $|\pm\rangle$

The conversion that must be automatic:

$$\begin{aligned} |+\rangle &= \frac{|0\rangle + |1\rangle}{\sqrt{2}}, & |-\rangle &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ |0\rangle &= \frac{|+\rangle + |-\rangle}{\sqrt{2}}, & |1\rangle &= \frac{|+\rangle - |-\rangle}{\sqrt{2}} \end{aligned}$$

It is its own inverse — the same formula both ways. In general for $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$:

$$c_{\pm} = \langle \pm | \psi \rangle = \frac{\alpha \pm \beta}{\sqrt{2}}, \quad |\psi\rangle = c_+ |+\rangle + c_- |-\rangle.$$

($c_k = \langle E_k | \psi \rangle$ means *conjugate* the eigenvector's entries first.)

Step 3 — stamp the phases, then read the answer

Worked archetype (Q41-type, 8 marks)

$$H = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \quad |\psi(0)\rangle = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{2/3} \end{pmatrix}, \quad t = \pi.$$

- $a = 1, b = \frac{1}{2}$: $E_+ = \frac{3}{2}$ on $|+\rangle$, $E_- = \frac{1}{2}$ on $|-\rangle$.
- $c_{\pm} = \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{3}} \pm \sqrt{\frac{2}{3}} \right)$, i.e. $c_+ \approx 0.986$, $c_- \approx -0.169$.
- $|\psi(\pi)\rangle = e^{-i\frac{3}{2}\pi} c_+ |+\rangle + e^{-i\frac{1}{2}\pi} c_- |-\rangle = i c_+ |+\rangle - i c_- |-\rangle = i(c_+ |+\rangle - c_- |-\rangle) \hat{=} c_+ |+\rangle - c_- |-\rangle$.

Back in the computational basis that is $iX|\psi(0)\rangle = i \begin{pmatrix} \sqrt{2/3} \\ 1/\sqrt{3} \end{pmatrix}$ — the amplitudes simply swap. Global phase i : state the equivalence, then drop it.

Same thing via the operator (faster when you must name a gate)

$H = aI + bX$ with $X^2 = I$ gives

$$e^{-iHt} = e^{-iat} (\cos(bt) I - i \sin(bt) X).$$

At $t = \pi$: $e^{-i\pi} (\cos \frac{\pi}{2} I - i \sin \frac{\pi}{2} X) = (-1)(-iX) = iX$. So the gate implemented is X (up to global phase) — that is the Q42 follow-up.

General rule for any H with $H^2 = I$ (e.g. X, Z , and $\begin{pmatrix} 0 & e^{-i\varphi} \\ e^{i\varphi} & 0 \end{pmatrix}$):

$$e^{-iHt} = \cos t I - i \sin t H.$$

Two more archetypes

- Diagonal** $H = \text{diag}(1, 2)$, $t = \pi$: $U = \text{diag}(e^{-i\pi}, e^{-2i\pi}) = \text{diag}(-1, 1) = -Z \hat{=} Z$.
- $\tilde{H} = \begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}$ on $|+\rangle$, $t = \pi$: $|+\rangle$ is an eigenstate with $E = a + b = 1$, so $|\psi(\pi)\rangle = e^{-i\pi} |+\rangle = -|+\rangle \hat{=} |+\rangle$. One eigenvalue, no expansion needed.
- Ground state of** $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: eigenvalues 2, 0; ground state = eigenvector of the *smallest* energy $= |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Reading off the gate, up to global phase

Pull out the first nonzero entry as a factor c with $|c| = 1$; compare what is left with

$$I, X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}, H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$-Z$, iX , $-iX$ are physically Z , X , X . Say so explicitly — the notes do it every time, and it is usually worth a mark.

Free sanity checks

- $\sum_k |c_k|^2 = 1$ before evolving, and $|c_k|$ never changes with t — time evolution only rotates phases. If a magnitude moved, you slipped.
- $|\psi(t)\rangle$ is still a unit vector. U_t is unitary.
- $t = 0$ must give back $|\psi(0)\rangle$.
- Energy is conserved: $E_{\text{tot}} = \langle \psi | H | \psi \rangle = \sum_k |c_k|^2 E_k$, same at every t .
- Probabilities of measuring an *energy* eigenstate are constant; probabilities in the computational basis are not (that is the whole point).

Your five recurring leaks

1. “A 2×2 has one eigenvalue.” It has two. Always solve the full quadratic.
2. Sign in $H - \lambda I$ for negative λ : the diagonal *increases*.
3. Reporting $v = 0$ from the null space. Trivial solution $\neq$ eigenvector.
4. Algebra under time pressure: $\frac{2}{\sqrt{2}} = \sqrt{2}$, $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$, $e^{-i\pi} = -1$, $e^{-i\pi/2} = -i$, $e^{-2i\pi} = 1$.
5. **Finishing.** Three of your last four lost marks were unfinished answers, not wrong maths. Re-read the question: normalized? every sub-part? global phase mentioned? ground state *stated as a vector*?

Cover this column and answer

1. Eigenpairs of $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$?
2. $|1\rangle$ in the $|\pm\rangle$ basis?
3. $H = \text{diag}(\pi, 2\pi)$, $t = 1$: which gate?
4. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ acting on $|0\rangle$ for time t : give $|\psi(t)\rangle$.
5. Ground state of $\begin{pmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{pmatrix}$?

1. 3 on $|+\rangle$, 1 on $|-\rangle$. 2. $(|+\rangle - |-\rangle)/\sqrt{2}$. 3. $\text{diag}(e^{-i\pi}, e^{-2i\pi}) = \text{diag}(-1, 1) = -Z \hat{=} Z$. 4. $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$, $E_{\pm} = \pm 1$, so $|\psi(t)\rangle = \frac{1}{\sqrt{2}}(e^{-it}|+\rangle + e^{it}|-\rangle) = \cos t|0\rangle - i \sin t|1\rangle$. 5. $E_+ = 1 < E_- = 2$, so $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.